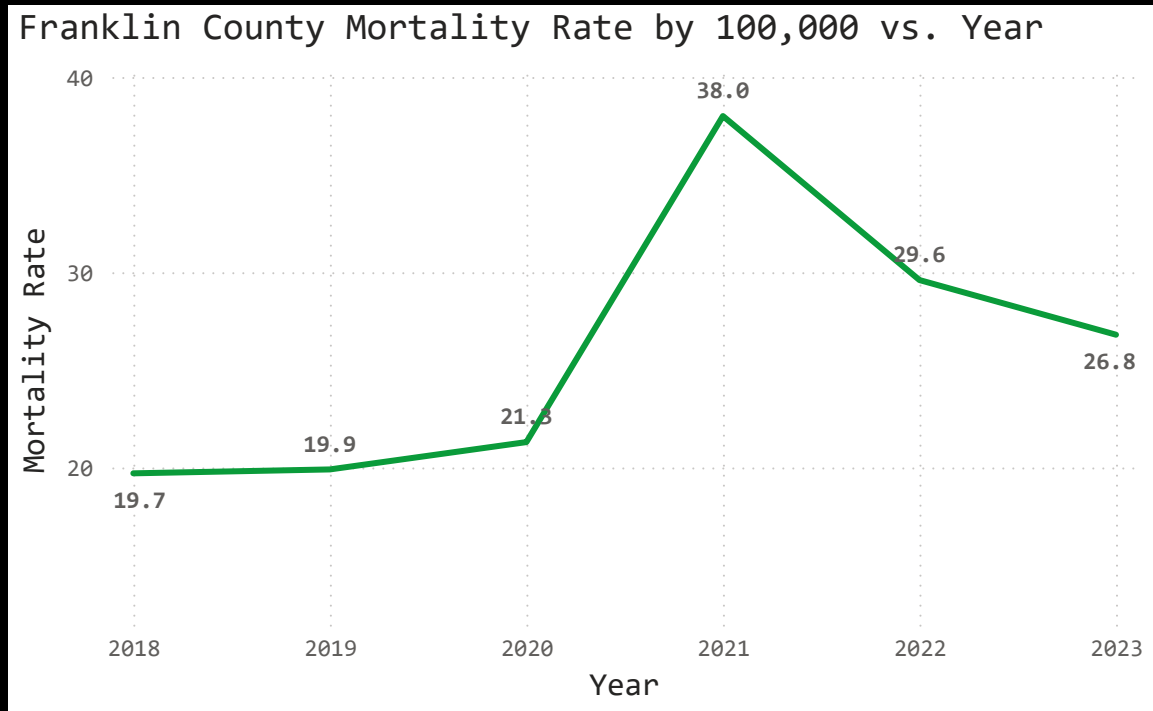
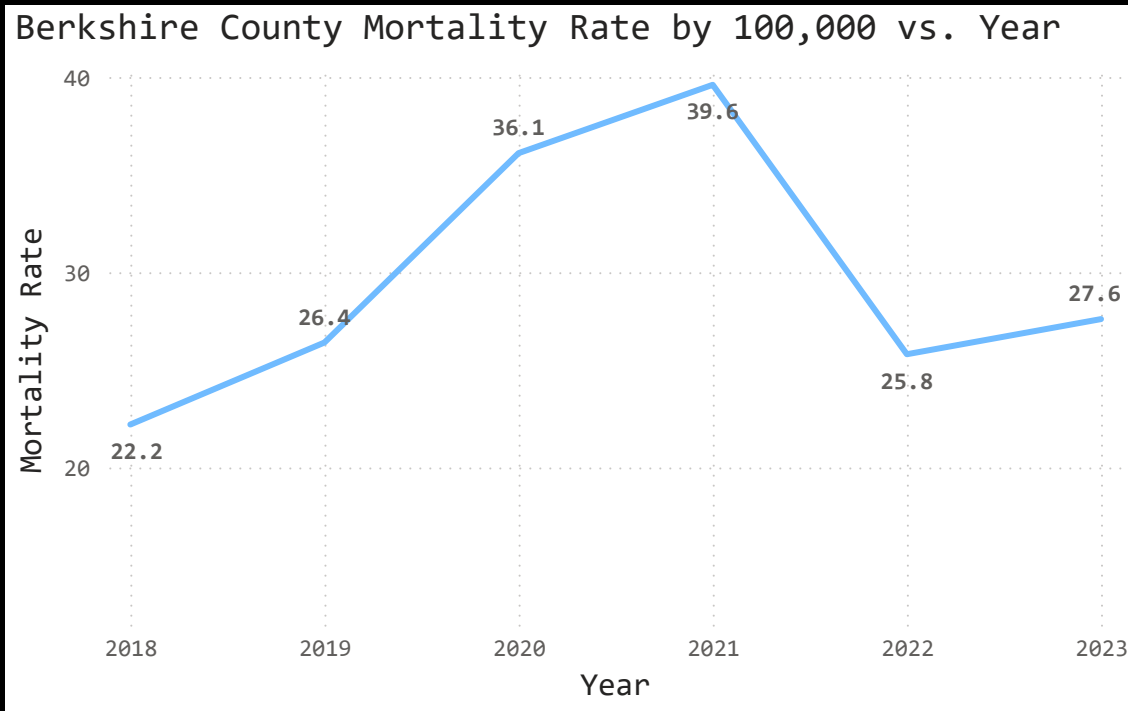
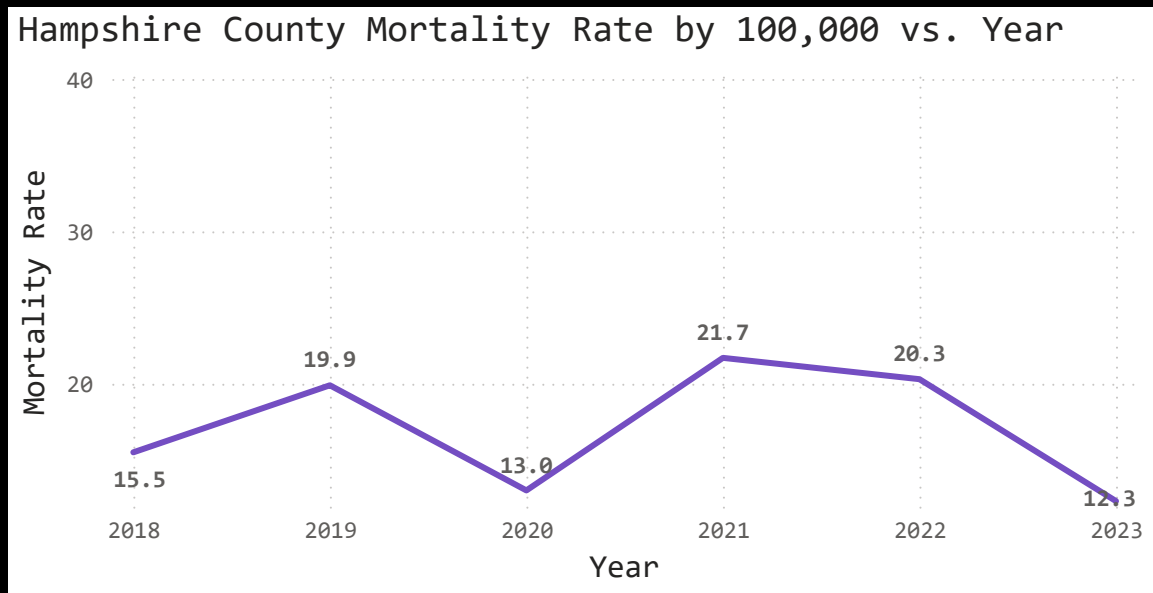
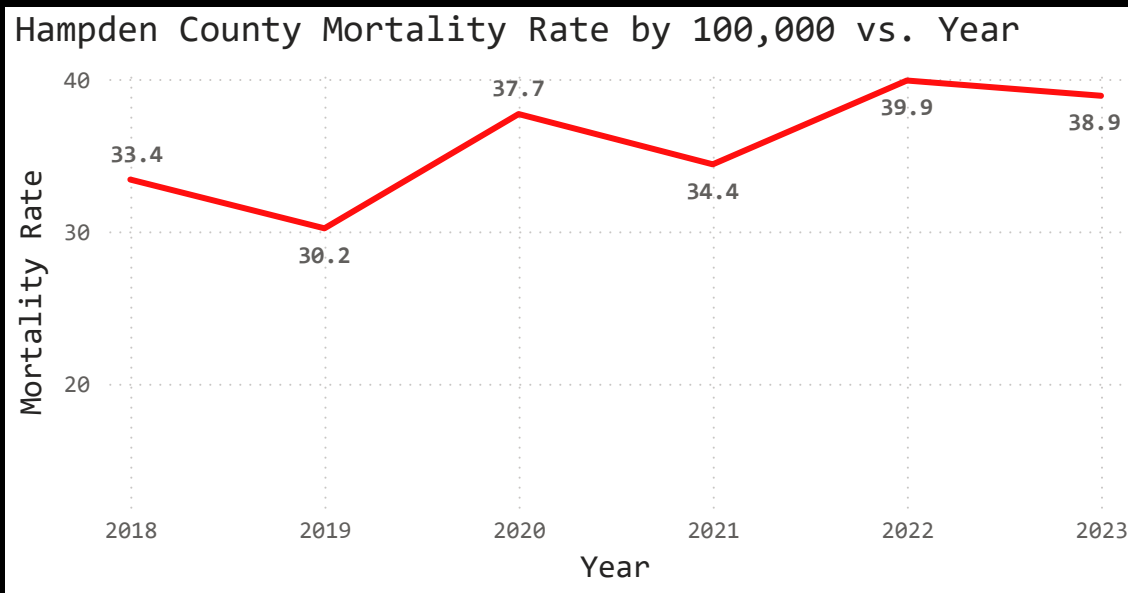


Opioid Mortality Rate by County in Western Massachusetts

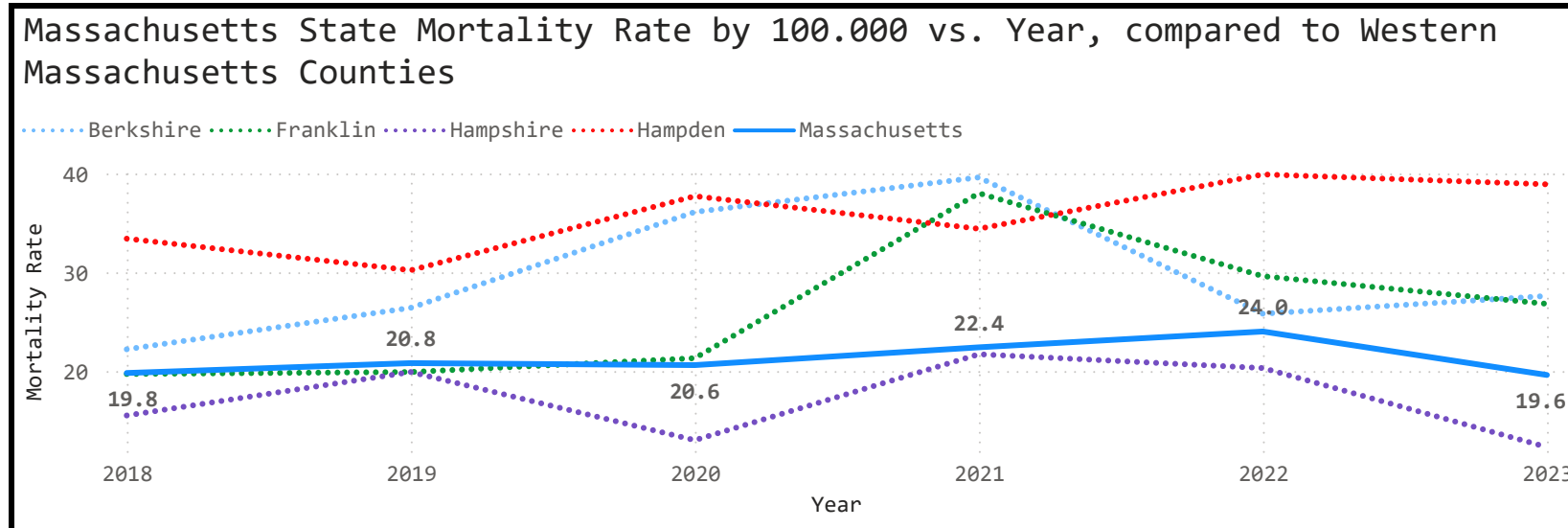


Death due to ICD-10 Code X42 is used to calculate the dependent variable.



Data is collected from wonder.cdc.gov.

Opioid Mortality Rate by County in Western Massachusetts (cont.)

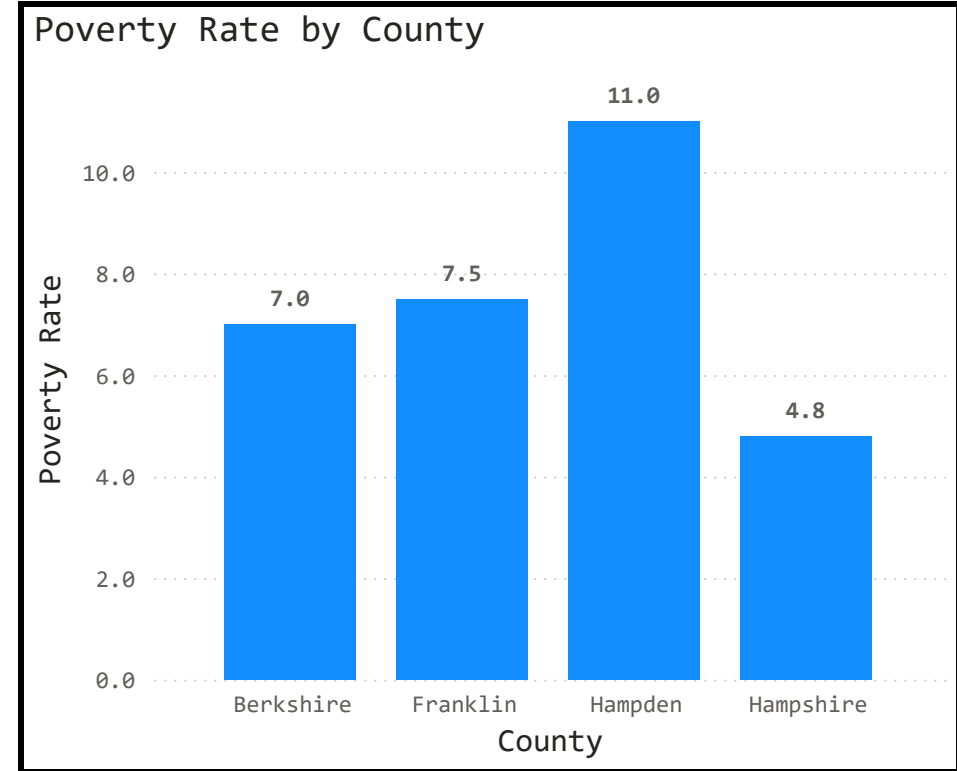
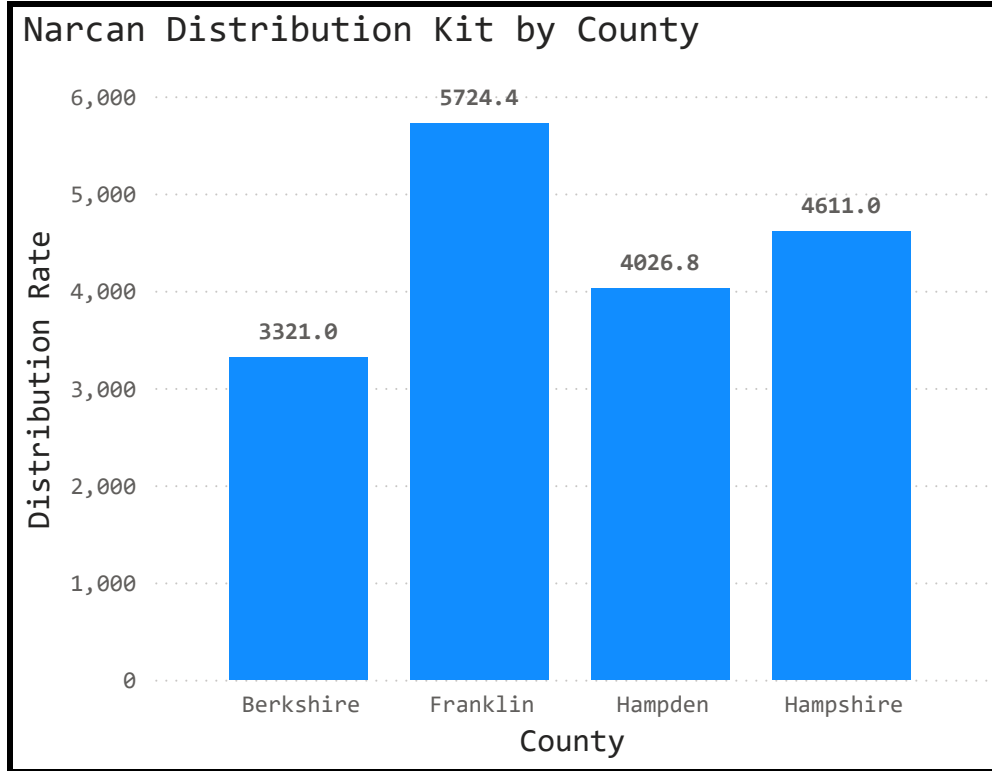


Based on this chart comparing overall state opioid mortality deaths per 100,000 people to counties in Western Massachusetts, we can observe that counties in Western Massachusetts, with the exception of Hampshire County, consistently suffer higher mortality rates than the state average. It is worth noting that while CDC data is available up until 2024, the data I have chosen to use only targets the 5 years between 2018-2023. This is due to Franklin County's 2024 X42 mortality data being unavailable, potentially due to the amount of deaths being under 10 for that year, resulting in that information being kept confidential.

We also observe here that in 2021, most counties faced a spike in X42 mortality rates. The cause of this increase is well-documented, as local drug supply was greatly contaminated by fentanyl saturation. This, along with still coming off of hindered healthcare access due to COVID-19, are supported reasonings for this increase. Massachusetts also serves as a distribution hub for illegal substances.

Hampshire and Hampden County are two bordering neighbors, yet throughout these 5 years, one consistently out-performs the other.

Opioid Mortality Rate by County in Western Massachusetts (cont.)



Although looking at Narcan Distribution Rates does not provide an explanation as to why Hampden and Hampshire County have contrasting results, something else we can look at are socioeconomic factors. Hampshire County has the lowest poverty rate for counties in Western Massachusetts. Characteristics of Hampshire County, such as being home to the Five College Area and its main city, Northampton, which drives the area to having a stable and growing economy, are potentially the two biggest factors here.

Much of Hampshire County being the center for college towns makes this reasoning hard to interpret based off data alone. What is commonly known though, is that wealthier, highly educated populations have better access to quality healthcare and are less likely to abuse substances.

Franklin and Berkshire County see similar poverty rates, and similar mortality rate trends. Franklin County's Narcan Distribution Rate being the highest, yet still facing high X42 mortality rates, indicates further policy intervention may be needed.